# Palo Alto Firewall
# URL Filtering

## 1. Objective

Configure URL filtering on a Palo Alto Firewall to block access to specific social media websites (YouTube and Facebook) using a custom URL category and applying it through a URL filtering profile attached to a security policy.

## Lab Environment

**Virtualization Platform:** VMware Workstation
**Firewall Version:** PAN-OS 11.2.0
**Firewall Management IP:** 192.168.1.15
**Client Machine:** Ubuntu 24.04 LTS
**Client IP Address:** 192.168.2.1
**Test Browser:** Brave

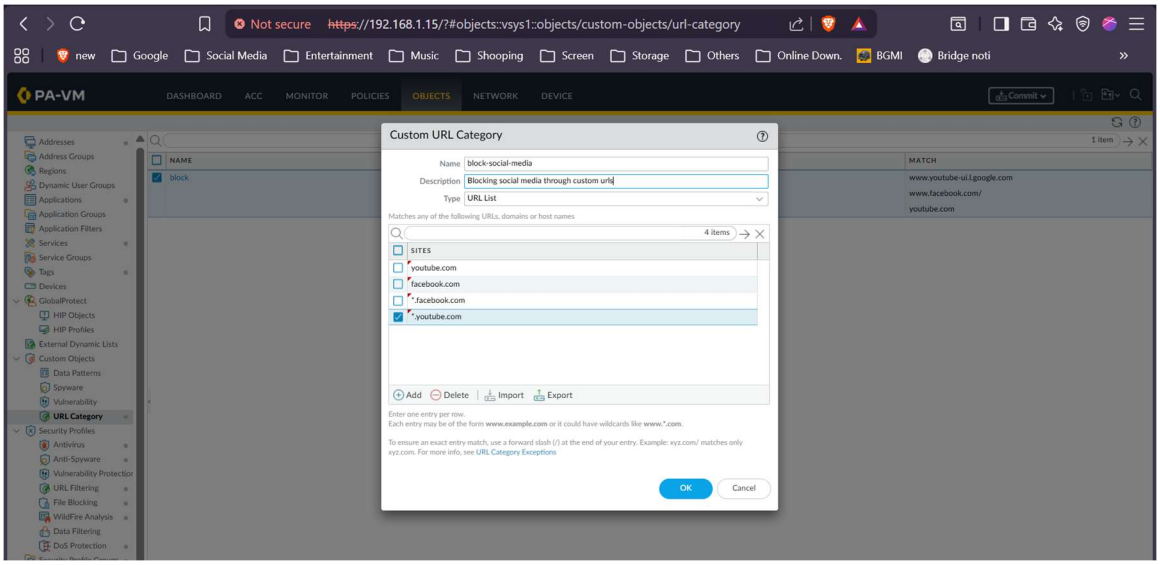
## Network Topology Diagram

The below diagram represents the lab architecture where the Ubuntu VM connects to the internet through the Palo Alto Firewall.

## 2. Configuration Steps

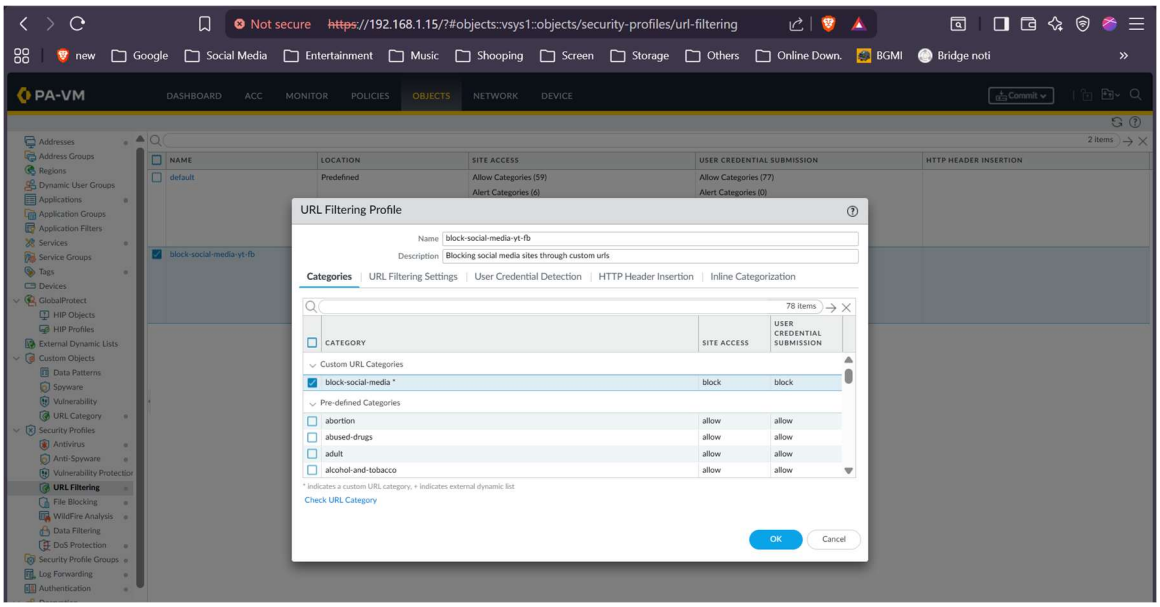
### Step 1 – Create Custom URL Category

Navigate to Objects → Custom Objects → URL Category. Create a category named 'block-social-media' and add the following URLs: youtube.com, *.youtube.com, facebook.com, *.facebook.com.



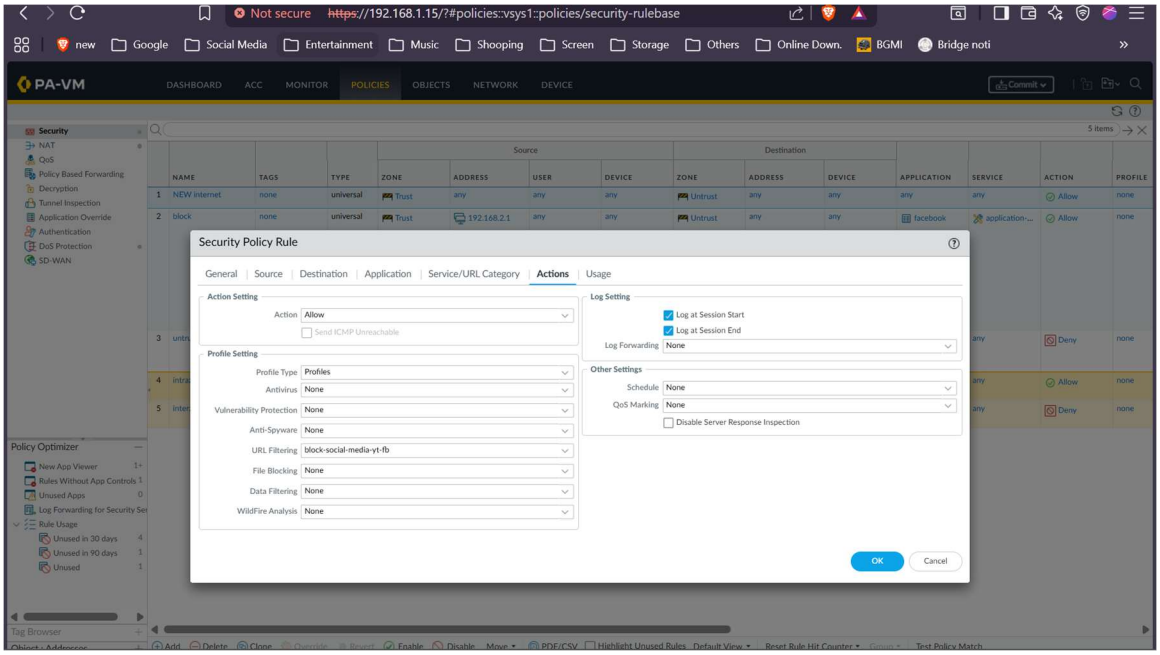

### Step 2 – Create URL Filtering Profile

Navigate to Objects → Security Profiles → URL Filtering and create a profile named 'block-social-media-yt-fb'. Under Custom URL Categories set the action for the category 'block-social-media' to BLOCK.



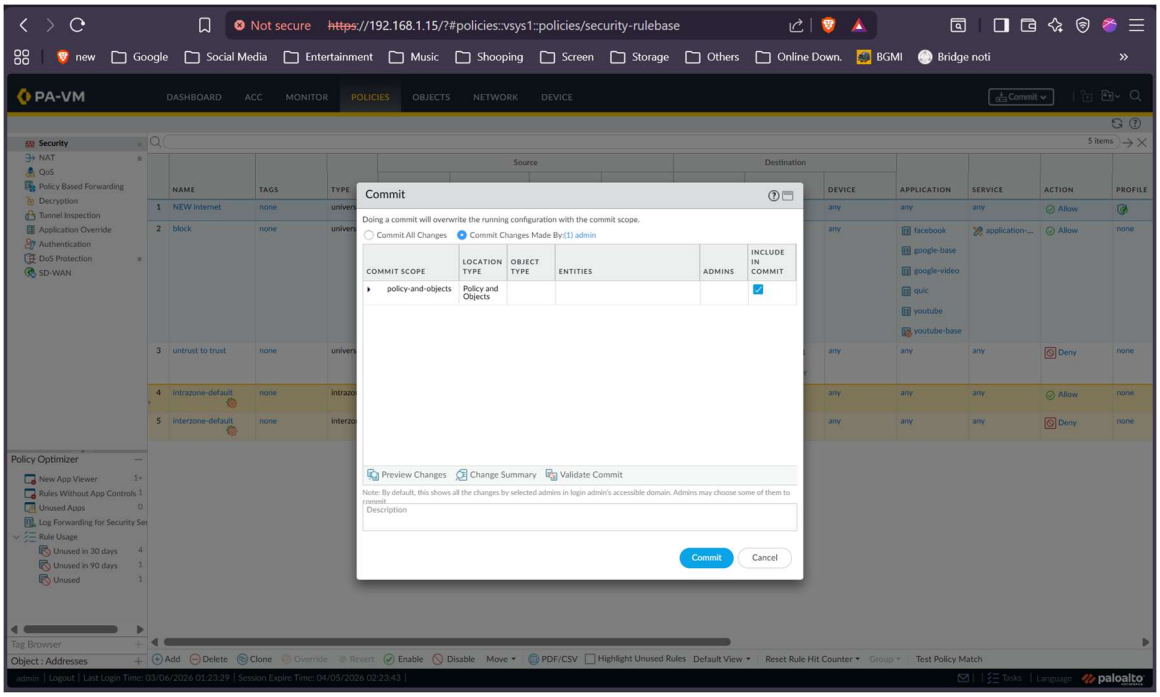
## Step 3 – Attach URL Filtering Profile to Security Policy

Navigate to Policies → Security. Edit the security rule allowing traffic from Trust to Untrust and in the Actions tab attach the URL Filtering profile 'block-social-media-yt-fb'.



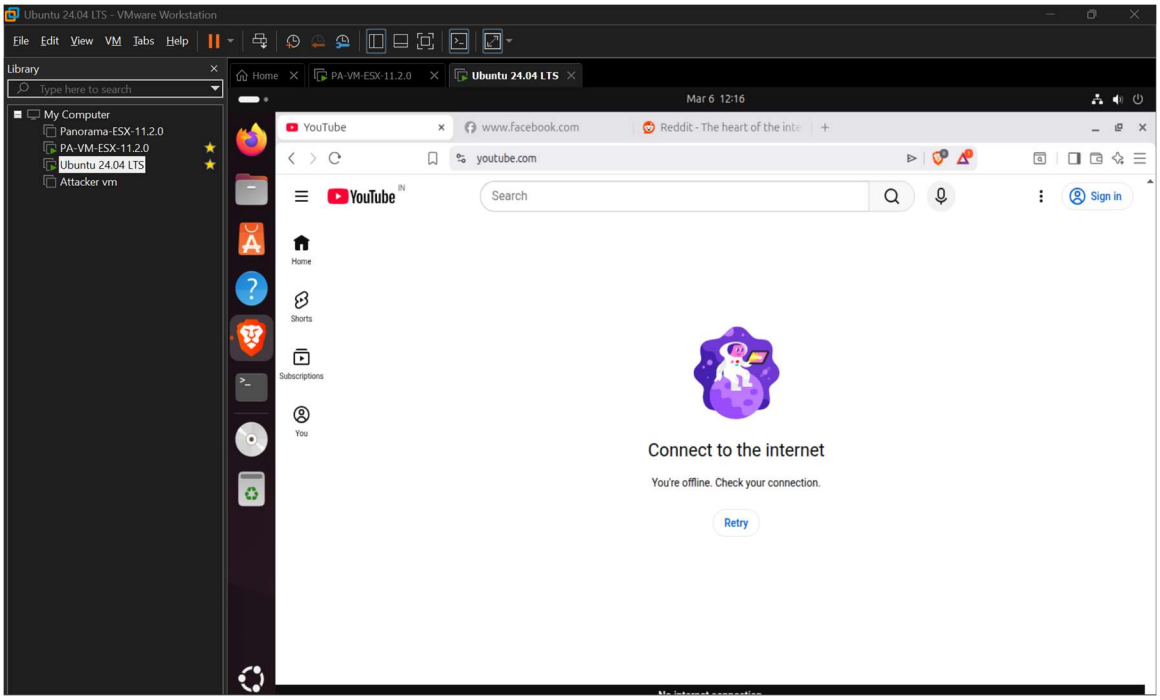

## Step 4 – Commit Configuration

Click Commit in the top-right corner of the firewall interface to apply the configuration changes to the running configuration.



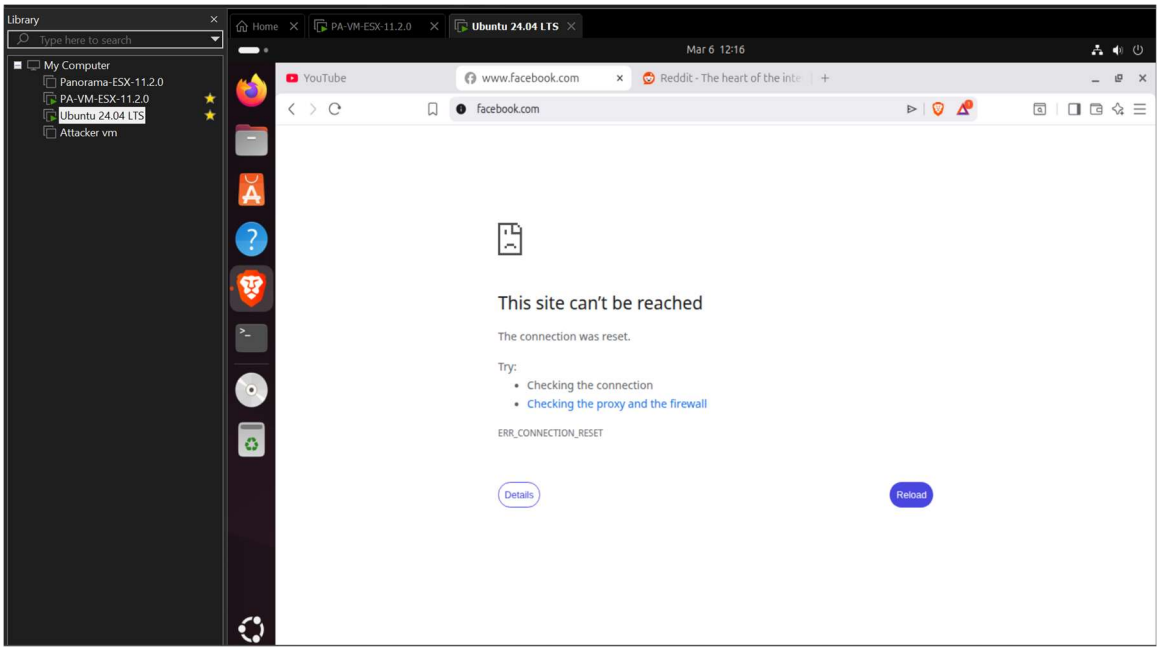
## Step 5 – Test Blocking of YouTube

From the Ubuntu client machine open a browser and try accessing youtube.com. The website should fail to load because the firewall blocks it.



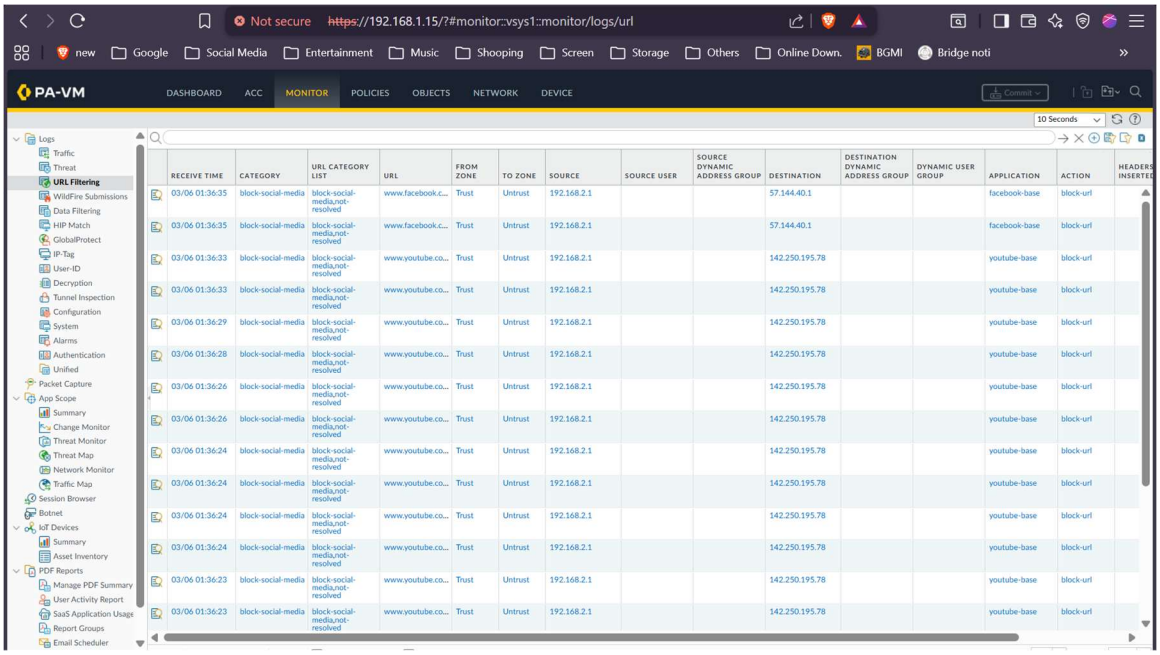

## Step 6 – Test Blocking of Facebook

Open facebook.com from the client machine. The firewall should reset the connection, indicating the site is blocked.



## Step 7 – Verify Logs in Firewall

Navigate to Monitor → URL Filtering to verify that access attempts to YouTube and Facebook are logged and blocked by the configured policy.




## 4. Result

The Palo Alto firewall successfully blocks access to YouTube and Facebook from the client machine. Attempts to access these sites are logged in the URL Filtering logs with the action 'block-url'.

## 5. Conclusion

This lab demonstrates how Palo Alto URL filtering can be implemented to control web access within a network. By creating a custom URL category and applying it through a URL filtering profile in a security policy, administrators can enforce web usage policies effectively.

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